



Infection Prevention and Management Manual

Specimen Collection for Microbiological Examination Guideline

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Hospital Wide	Clinical Staff

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Introduction

This guideline applies to specimens sent for microbiological examination.

Correct specimen collection labelling, patient identification and completion of request form are essential to ensure timely, accurate results that aid in diagnosis and treatment.

Laboratory specimen processing can only occur when all the following requirements are met:

- Request form completion;
- Correct patient identification; and
- Correct specimen labelling.

Responsibilities

The medical officer requesting the specimen is responsible for completing the request form or the Computerised Provider Order Entry (CPOE).

All collection equipment must be within expiry date as collection equipment past its expiry date can affect the integrity of the sample.

Appropriate Personal Protective Equipment (PPE) must be worn when collecting specimens as per standard precaution requirements.

Use aseptic technique for the collection of specimens.

Care should be taken not to contaminate the outside of the container or the request form.

Ensure lids/tops are firmly in place to avoid leakage.

The person collecting the specimen is responsible for completing the collection details on the request form and labelling the specimen container prior to transport.

Use of Pneumatic Tube

When sending glass blood culture bottles via the pneumatic tube, an appropriate plastic bottle holder/container **MUST** be used.

The following microbiological specimens must not be sent via the pneumatic tube:

- Specimens on ice
- Specimens from patients with known or suspected Viral Haemorrhagic Fever or Severe Acute Respiratory Pathogens, e.g., MERS / avian influenza / SARS (excluding COVID-19)

Specimen requirements

Clearly label the specimen with patients' full name, unique medical record number (UMRN), date of birth, time, date and anatomical site of collection. The patient's label should be used whenever available.

Full name and date of birth or full name and UMRN are minimum requirements to avoid compulsory rejection by the laboratory.

If the specimen is urgent, please state clearly "URGENT" on request form.

The specimen must be separated from the form in the dual compartment plastic bag, prior to transport. Place the specimen in the sealed compartment. Larger specimens should be sent in a plastic bag and secured.

All specimens should be sent to:

- **SCGH:** Central Specimen Reception Area (CSRA), PathWest Ground Floor, PP Block, as soon as possible after collection. CSRA is open 24 hours per day, 7 days per week.
- **OPH*:** OPH Pathology Collection, D Block
Monday to Friday 0700-1700hrs (excluding public holidays)
Saturday 0700-1200hrs
Sunday and public holidays 0700-1100hrs
* For A/hrs urgent microbiological specimens contact the OPH After-hours Clinical Nurse Manager who will prioritise for courier to SCGH CSRA as required.

Specimens sent for bacterial examination should be collected before antimicrobial therapy is commenced where possible. If this is not possible, please make a note of this on the pathology request.

General specimen collection equipment is ordered through the ward imprest. Additional and specialised equipment can be obtained by contacting:

- **SCGH:**
 - Viral transport medium (VTM) PathWest Central Specimen Reception Area (CSRA), ground floor PP block on Ext. 34061
 - Mycolytic F bottles – TB Laboratory PP Block on Ext. 34561
- Corneal scrape agar plates – Bacteriology PP Block on Ext. 34564
- **OPH:** PathWest Collection, D Block on extension 78088.

For patients with a recent international travel history, where there is a suspicion of an exotic disease, e.g. *Brucella* spp., *Histoplasma* spp., there may be considerations for special handling of specimens relevant to Risk Group 3 or 4 organisms. Please contact the clinical microbiologist before collecting the specimens and indicate the suspected infection on the request form.

For specimen collection for suspected Creutzfeldt-Jakob Disease (CJD), refer to [Infection Prevention and Management Manual Creutzfeldt-Jakob Disease Risk Assessment and Management Policy](#).

For specimen collection for suspected Viral Haemorrhagic Fever, refer to [Infection Prevention and Management Manual Viral Haemorrhagic Fever Policy](#).

Table 1. Abbreviations

CAPD	Continuous ambulatory peritoneal dialysis
CPOE	Computerised Physician Order Entry
CRE	Carbapenem Resistant Enterobacteriaceae
CSF	Cerebrospinal Fluid
CSRA	Central Specimen Registration Area
CSU	Catheter Specimen of Urine
CT	Chlamydia trachomatis
CVC	Central Venous Catheter
HIV	Human Immunodeficiency Virus
HSV	Herpes Zoster Virus
ILI	Influenza-Like illness
MC&S	Microscopy, culture, and sensitivity
MERS	Middle East Respiratory Syndrome
MRSA	Methicillin Resistant <i>Staphylococcus aureus</i>
MSU	Midstream Specimen of Urine
NG	Neisseria gonorrhoeae
PCR	Polymerase Chain Reaction
PPE	Personal Protective Equipment
STI	Sexually transmitted infection
SARS	Sudden Acute Respiratory Syndrome
TB	Tuberculosis
UMRN	Unique Medical Record Number
UTM	Universal transport media
VRE	Vancomycin Resistant Enterococci
VTM	Viral transport media
VZV	Varicella Zoster Virus

Table 2. Specimens for Microbiological Examination

Specimen Type	Requirements/Specimen container	Procedure	Comments
Blood Cultures A blood culture set is defined as one aerobic bottle and one anaerobic bottle.	• Clean trolley or kidney dish/tray/receptacle • Personal Protective Equipment • Tourniquet • 70% sterile alcohol impregnated swab • Vacutainer system or 20mL syringe • Blood transfer device (e.g. butterfly) • Sharps disposal container • Blood Culture Bottles (two sets) • Small bandaid/plaster • Waste disposal bag Blood culture bottles are available from: <u>SCGH</u>- CSRA, PathWest, Ground Floor, PP Block, ext. 34061 <u>OPH</u>- PathWest Collection, D Block on extension 78088.	Follow standard aseptic technique. Obtain consent from patient, assess veins visually and clean patient's arm with soap and water if visibly soiled. 1. Perform hand hygiene 2. Clean trolley or kidney dish with pre-diluted disinfectant/ detergent wipe and allow to dry. 3. Gather equipment for the procedure. 4. Inspect packaging for damage, check sterility indicators and expiry dates. 5. Perform hand hygiene. 6. Prepare equipment using non touch technique. 7. Remove the caps from the blood culture bottles and scrub the vial septum for 30 seconds with 70% sterile alcohol impregnated swab. 8. Allow to dry.	Collect blood cultures as soon as possible after onset of clinical symptoms. Always take blood culture specimens at the beginning of sample collection series and inoculate blood culture bottles before any vacutainer tubes. Collect into the aerobic culture bottle before collecting into the anaerobic culture bottle. Ensure blood culture bottles have not exceeded expiry date. Take two sets of blood cultures from separate sites if patient is septic and therapy is to commence immediately. Fill each bottle with 8-10 mL of blood. • If blood is drawn through a central venous catheter (CVC) do not discard the first 5 mL of blood unless the CVC is antibiotic locked, then the first 5 mL of blood of blood should be discarded. • If blood is drawn through a central venous catheter, collect a second set of blood cultures from a peripheral vein at the same time.

		1. Label both bottles with the patient's label 2. Perform hand hygiene and prepare the patient 3. Apply tourniquet and identify a vein. 4. Perform hand hygiene. 5. Apply non-sterile gloves 6. Clean venepuncture site with 70% sterile alcohol impregnated swab 7. Allow to dry. Do not re-palpate the site. 8. Puncture vein 9. (a) If using vacutainer system, insert aerobic blood culture bottle into vacutainer holder. Allow 8-10 mL of blood to enter the blood culture bottle. Repeat with anaerobic blood culture bottle. (b) If using a blood transfer device and syringe - withdraw the 8-10 mL of blood required for each bottle (aerobic and anaerobic). If insufficient blood for both bottles (<16 mL), fill aerobic bottle first. Do not put less than 3.0 mL blood into any standard adult blood culture bottle. 10. Apply dressing using non touch technique. Release tourniquet 11. Dispose of sharps and waste, remove PPE, perform hand hygiene, clean the trolley.	
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Specimen Type	Requirements/Specimen container	Procedure	Comments
		12. Complete the request form, including the patient label, ensuring the <u>date and time</u> of collection together with the <u>site</u> of collection is documented.	
Cerebrospinal Fluid	Specimen containers to collect CSF available via PathWest Collection.	See Nursing Practice Guideline No. 17 Lumbar Puncture Management.	Collect into three sterile tubes. Increased volume (at least 10-15 mL) required if tuberculosis meningitis is suspected. Do not refrigerate tubes after collection.
Faecal Specimens Bacteriology/ Parasitology/ Virology	Personal Protective Equipment. Freshly sanitised bedpan / or clean macerator bedpan. Wooden spatula. Sterile yellow screw top container or dedicated brown screw top stool container.	Provide a freshly sanitised bed pan. Use a wooden spatula or dedicated brown top pot lid with scoop to transfer the specimen to the container. Faecal matter may be collected from incontinence pads.	Do not contaminate specimen with urine. Do not collect more than one specimen within a 24-hour period for MC&S. If more than one specimen is required, send consecutive daily specimens. Faecal specimens should be collected during the acute phase of the illness. When investigation for amoebae has been requested, the specimen should be conveyed to the laboratory for immediate examination while still warm.
Sexually Transmitted Infections (STIs) Chlamydia trachomatis (CT) Gonorrhoea Herpes simplex Syphilis Mycoplasma genitalium Trichomonas	Personal Protective Equipment. Orange top swab for PCR. Charcoal swab and glass slide for MC&S. Sterile yellow screw top container for INITIAL, *First pass urine* stream.	For chlamydia and gonorrhoea <u>Female</u> *First pass urine for Chlamydia trachomatis(CT) / Neisseria gonorrhoeae (NG) PCR - Clinician-collected endocervical swab (preferred) or self-collected vaginal swab for CT/NG PCR	When screening for STIs, serology for HIV, hepatitis B, hepatitis C and syphilis is also recommended.

Specimen Type	Requirements/Specimen container	Procedure	Comments
	First pass urine = the first 5-10mL of the urine stream. MSU is not suitable for STI PCR.	- Symptomatic: collect additional endocervical or vaginal swab for MC&S. <u>Male</u> *First pass urine for CT/NG PCR - If discharge is present, collect a urethral swab for MC&S Additional sites may need to be swabbed depending on symptoms and exposure history. See STIs - STI Guidelines Australia If herpes simplex or syphilis is suspected, collect separate swab from lesion for Polymerase Chain Reaction (PCR). Asymptomatic screening for Mycoplasma genitalium is not recommended.	
Skin Scrapings for fungal infections and nail clippings	Personal Protective Equipment. Sterile nail clippers/splitters Sterile blade Yellow screw top sterile container Sterile forceps Sterile scissors Sterile swab without transport medium Bluey.	Ensure the collection site is clean and dry: - Remove ointments/creams with alcohol wipe or sterile gauze soaked with sterile saline - Wash visibly soiled skin/nails with soap and water - Remove nail polish with acetone/nail polish remover. <u>Skin scrapings</u> - Holding skin taught, use a sterile scalpel blade and scrape the area at	The area to be scraped should be kept clean and dry. Collect different sites individually. Where possible, topical treatment should cease two (2) days prior to specimen collection.

Specimen Type	Requirements/Specimen container	Procedure	Comments
		a 45° angle to remove the outer surface of the skin. Scrape across the lesion, paying attention to the growing margin for 30 seconds to a couple of minutes. • If blisters are present, excise the top of the blister and add this to the scrape container. For moist areas and no/or only a small collection obtained, collect a sterile swab moistened with sterile saline and vigorously rub the area of collection. <u>Nail clippings</u> • Using nail clippers/splitters, make small 'V' shaped cuts, no more than 2mm at a time, only clipping deep enough to expose the crumbly material at the nail bed. • If nails are too short to clip, use a scalpel blade to file the affected parts of the nail • Use scalpel blade, collect crumbly material, white patches, discoloured lined or other debris located at the top of the surface nail into the specimen collection container • Collect a sterile swab with, and without transport media of any exudate present	

Specimen Type	Requirements/Specimen container	Procedure	Comments
Sputum for Microscopy, Culture and Sensitivity, (MC&S)	Personal Protective Equipment. Sterile, yellow screw top container.	Early morning collection is preferred, before food, especially for tuberculosis. Rinse mouth with water prior to collection. Ask patient to cough, not spit.	Specimens of saliva are inadequate for processing and will be rejected by the laboratory.
Sputum for tuberculosis	Personal Protective Equipment including Fit tested and fit checked P2/N95 respirator and eye protection. Single room required. Sterile, yellow screw top container.	Three specimens should be collected 8-24 hours apart. Samples should be collected early morning prior to eating and drinking as they have the greatest yield of organisms.	Specimens of saliva are inadequate for processing and will be rejected by the laboratory.
Induced sputum		See Induced Sputum Collection	
Swabs for COVID-19 / SARS-CoV-2, influenza, and other respiratory pathogens		For collection of nose and throat swabs, see Influenza-like Illness Policy	
Lesion swabs for PCR, e.g. HSV / VZV	Personal protective equipment. Dry swab for PCR (orange top) or flocked swab with VTM. NOTE: Only dry swabs (dry = no media) or swabs in VTM are suitable for PCR.	If a vesicle is present, de-roof the vesicle and use the swab to collect the underlying fluid. Gently rub/roll the swab against the base of the lesion. Place swab back into holder or into VTM.	

Specimen Type	Requirements/Specimen container	Procedure	Comments
Rapid Antigen Test (RAT)	Personal protective equipment which includes fit tested and fit checked P2/N95 respirator. Rapid Antigen Test (RAT) Contact and airborne precautions.	For Acute Respiratory Illness / COVID like symptoms	Please refer to manufacture instructions
Swabs Microscopy, Culture and Sensitivity (MC&S) e.g. wound swab	Personal Protective Equipment. Swab collection kit, available on ward imprest.	Irrigate wound with sterile saline prior to swabbing to eliminate surface contaminants. Slide for Microscopy. • Use plain swab (lightly moistened with sterile saline if wound is dry) • Rotate the wound swab over a 1 cm² area of the wound • Gently roll the used swab across the centre of the glass slide • Allow one (1) minute to air dry • Return glass slide to holder, label specimen container • Discard used swab. Swab for Culture • Use swab with medium (lightly moistened with sterile saline if wound is dry) • Rotate the swab over a 1 cm² area of the wound • Insert swab into medium.	Always send pus/exudate rather than a swab if possible. Collect in a syringe and transfer to a sterile yellow screw top container. For cavity and sinus wounds take swab from the deepest point. For wound swabs, ensure the location of the wound is documented on the pathology request. (e.g., left shin ulcer, right hip surgical wound, sacral pressure injury, right thumb dog bite puncture). Swabs may be pre-moistened with a small amount of sterile saline.

Specimen Type	Requirements/Specimen container	Procedure	Comments
Specimens for multi-resistant organism (MRO) screening, including		Includes MRSA, VRE, carbapenemase-producing organisms (CPO), <i>Candida auris</i> . See Multi-resistant Organism Policy	
Throat swabs For MC&S and PCR	Personal Protective Equipment Tongue Depressor and torch Sterile swab in charcoal transport media MC&S for bacteriology. Dry (no media) sterile swab for PCR (white or orange top swab). Plastic shafted swab and viral transport media (VTM) for viral culture.	Use a tongue depressor to hold tongue down. Carefully but firmly rub the swab over tonsils and posterior pharynx. If there are any areas of exudate, collect a specimen of the exudate. Do not touch cheek, teeth or gums as the swab is withdrawn from the mouth. Insert swab into transport medium. If specimen is to be sent for virology use a dry (no media) plastic shafted swab for PCR placed in virus transport medium. (Orange or white top swab).	Please ensure virology and PCR specimens are placed into the VTM. Virology and PCR cannot be performed if specimen is placed in MC&S transport media (i.e., charcoal).
Urine Collection - Mid Stream Urine (MSU)	Personal Protective Equipment. Sterile kidney dish (obtained from SSD / part of catheter packs). Freshly sanitised urinal or bedpan (if required) - this would be considered a clean catch and should be documented on the request form.	Wash genital area with soap and water then dry. <u>Female</u> : wash from front to back. <u>Male</u> : if uncircumcised, retract foreskin, clean meatal area around head of penis. 1. Void first part of urine into toilet/bedpan. 2. If the patient is able: stop urine flow momentarily; collect a portion of urine into a sterile yellow screw top container. If patient cannot do this,	Be careful not to touch inside or lip of container. If only a 'clean specimen' can be obtained, please indicate this on the laboratory request form.

Specimen Type	Requirements/Specimen container	Procedure	Comments
	Urine collection device or sterile yellow screw top container.	use a sterile kidney dish to collect a specimen and transfer into a sterile yellow screw top container.	
Urine Collection - Catheter Specimen Urine	Personal Protective Equipment. Sterile alcohol swab 10 mL syringe Catheter clamp Sterile yellow screw top container.	• Clamp off drainage tubing below sampling port • Swab sampling port on the catheter tubing with alcohol swab, allow to dry. Insert syringe and withdraw specimen – 10 mL is required by laboratories • Remove syringe from sampling port • Carefully transfer urine into sterile yellow screw top container • Dispose of syringe • Unclamp catheter.	For catheterised patients, specimen collection for microbiological culture should be avoided unless the patient has systemic signs (S&S) or symptoms of infection. If the patient has S&S of an infection and the IDC has been in for >14 days, the IDC should be removed, replaced and then a sample collected at the time of insertion of a new IDC. Specimens should only be collected through the designated sampling port on the urine drainage tubing. NEVER from the collection bag.
Urine Collection Tuberculosis	Personal Protective Equipment Large sterile container X 3	• An early morning specimen is required on three (3) separate days • Place <u>total morning void volume</u> into sterile container. A minimum amount of 30mls is required to process urine for tuberculosis testing. Note: a separate collection container is required for each collection day. • Place container in large plastic bag.	
Urine Collection from Ileal and Colonic Diversion		See SCGOPHCG Nursing Practice Guideline, Colostomy Ileostomy Urostomy.	

Related Documents

National Standard/s

- Australian Commission in Safety and Quality in Health Care. National Safety and Quality Health Service Standards. Standard 3 Preventing and Controlling Healthcare Associated Infections.

Related Documents

- [SCGOPHCG Infection Prevention and Management Manual:](#)
 - Multi-Resistant Organism (MRO)
 - Influenza-Like Illness Policy
 - Aseptic Technique Policy
 - COVID-19 Policy
 - Tuberculosis Policy
 - Varicella.
- [SCGOPHCG Nursing Practice Guideline:](#)
 - Lumbar Puncture Management
 - Colostomy Ileostomy Urostomy
 - Bladder Management and Urinary Catheterisation.

Bibliography

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<http://www.hpa.org.uk/ProductsServices/MicrobiologyPathology/UKStandardsForMicrobiologyInvestigations/TermsOfUseForSMIs/AccessToUKSMIs/SMIBacteriology/>

Authorisation & Version Control

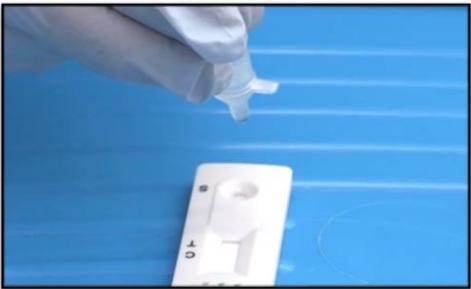
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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form.

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1: Microbiology Specimen Collection Equipment

EQUIPMENT	SPECIMEN
	Aerobic and Anaerobic Bottles: Used for Blood Cultures Always use an aseptic non touch technique. Check expiry dates and use both bottles for aerobic and anaerobic bacteria. Never overfill bottles (8-10mls is required in each bottle)
	Brown Top Pots: Used for Stool Specimens Used for microbiology, virology and parasite faecal examination. Do not contaminate specimen with urine. For amoeba investigation the sample must be sent to the laboratory immediately.
	Yellow Top Pots: Used for Sputum, Mid-Stream Urine and body fluid specimens (except CSF, CAPD fluid) The yellow top pots are used for sputum and urine but can also be used for pus and tissues for culture. Sputum specimens are best collected early in the morning, prior to eating and drinking.
	Dry Swabs (No media) Swabs without charcoal media are to be used for VRE and CPO (both rectal swabs, moistened with sterile saline). These swabs can also be used for MRSA screening (nose, throat and wounds). Moisten swab with sterile saline if the area being swabbed is dry.
	Charcoal Swabs Charcoal media should be used for wound, throat, eye, ear, skin and genital swabs (for MC&S) and can be used for MRSA screening (nose, throat and wounds). Moisten swab with sterile saline if the area being swabbed is dry. NOT to be used for VRE/CPO specimens.
	Metal or Plastic Shafted Dry Swabs: Used for Viral Specimens for PCR and culture The swab end is to be placed into the viral transport medium (VTM), This is stored in the fridge. Examples include influenza (ILI) nose/throat swabs and herpes/shingle lesion swabs. Send to laboratory immediately.

	Viral Transport Medium (VTM) VTM is to be stored in the fridge. Check expiry date prior to use. Place the swab tip from viral specimens for PCR into the VTM by cutting it with a pair of sterile scissors and send to the laboratory immediately. VTM is available on request via PathWest.
	Flocked swab Recommended for influenza like illness specimen collection. The swab and liquid universal transport medium (UTM) is in the same package and can be stored at room temperature. Check expiry date prior to use. Place swab into the UTM container, flocked end first, and snap the shaft at the indicated snapping point (red line), using the rim of the UTM tube for leverage if required of swab inside Close the UTM container with the flocked end of the swab inside, discarding the remainder of the shaft. Ensure that the UTM container is closed correctly and is not leaking – the swab is usually slightly bent within the tube.
	There are a variety of RAT kits available. • Follow the manufacturer's specific instructions, including quality controls • Store the kit in a cool dry area, between 2 and 30 degrees • If the test device is stored in a refrigerator, remove 30 minutes prior to testing to allow kit components to reach room temperature of 15 - 30 degrees Celsius